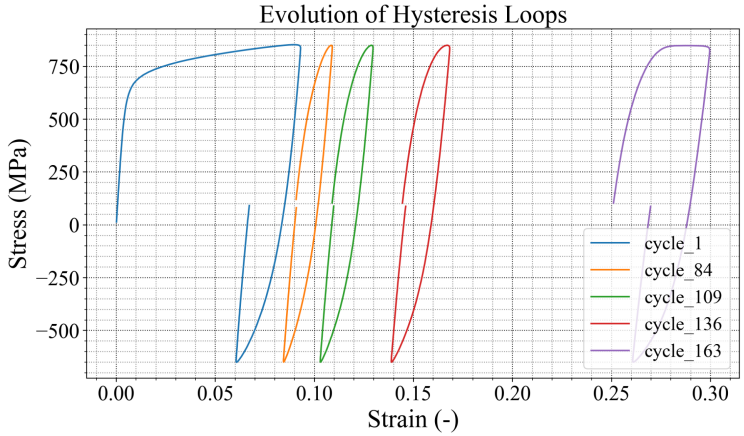
POST-PROCESSING RECORD

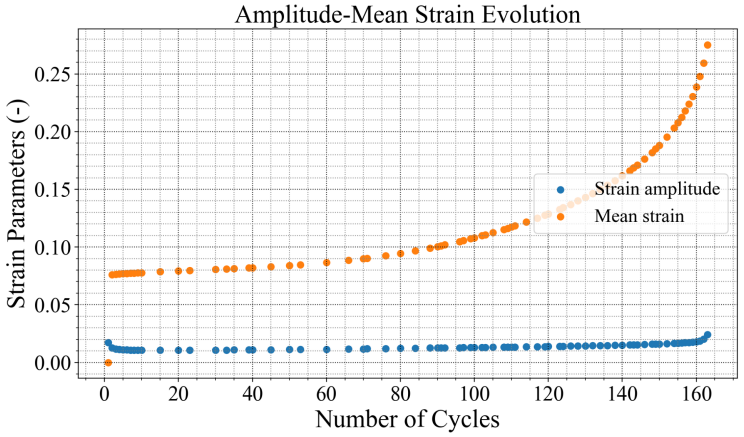
Title:	Ratcheting LCF test	Date:	2024-06-09
Specimen number:	ID59 solid	Dimensions:	diameter 6.01 mm
Material:	In718 SLM(Z)	Type of loading:	uniaxial
Equipment:	Instron 8801	Gauge length:	12.5 mm
Institution:	VUT Brno, Czechia	Experimenter:	Josef Zapletal
Temperature:	22°C	Controlled:	Stress-harmonic
Amplitude:	750 MPa	Mean value:	+100 MPa
Loading rate:	- kN/s	Unloading rate:	-
Frequency of load.:	2.0 Hz	Sampling frequency:	200 points/cycle

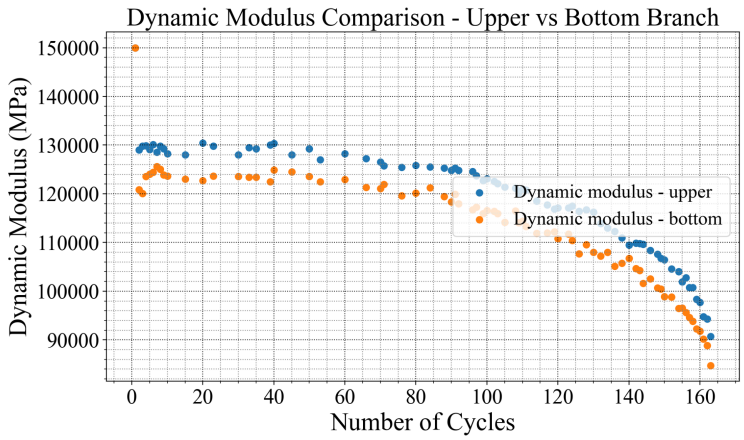
Input values during evaluation

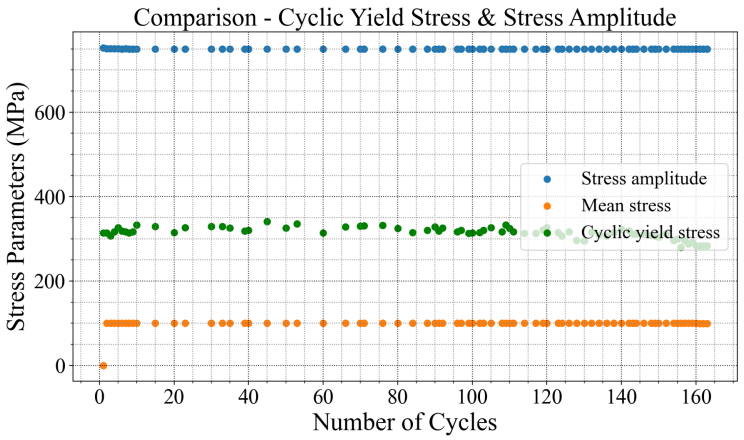
$N1_{RAT}$: 20 $N1_{E10}$: 70 Young modulus E: 150015 MPa El.region size $\Delta\sigma_{LIN}$: 200 MPa

$N2_{RAT}$: 40 $N2_{E10}$: 90 E - used for W_p: 150000 MPa Offset pl.strain: 5.00e-04









Half-life values:

ε_a (%)	ε_{ap} (%)	ε_{max} (%)	$d\varepsilon/dN$ (-)	N_f (-)	N_f^{E10} (-)	E_{dyn}^{upp} (MPa)	E_{dyn}^{btm} (MPa)	Y_{cyclic} (MPa)
1.2343	0.6238	34.8036	1.394e-04	164	144	125462.9	120136.6	320.87

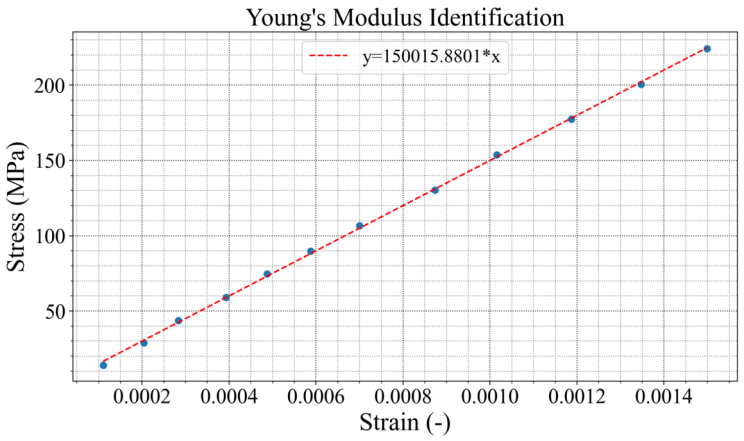
Note: Till fracture

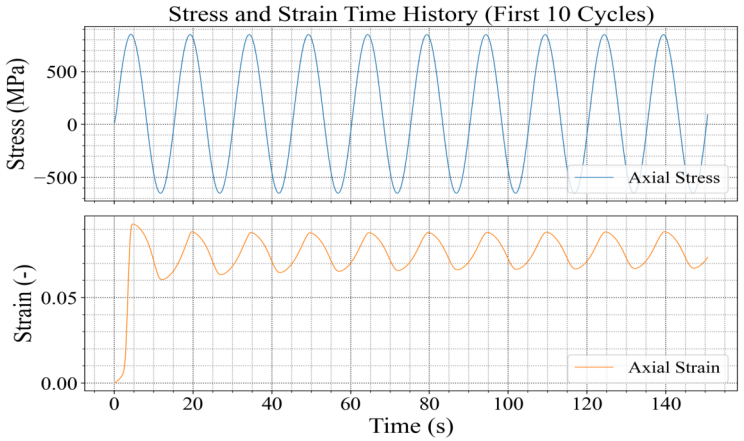
Acknowledgments: Financial support from the Czech Science Foundation under the registration no. 23-05338S.

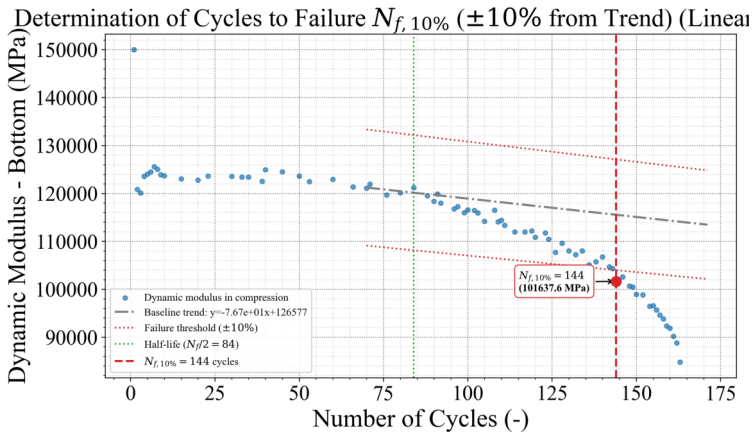
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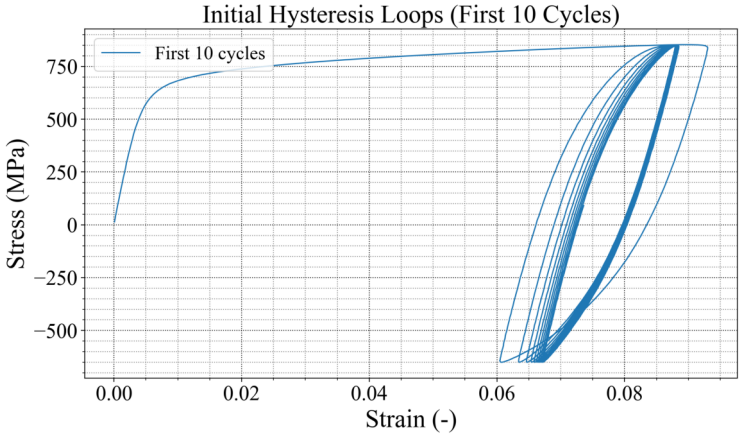
Ref: Tomáš Adamec et al. (2025) "Backstress shift modelling concept for improving uniaxial ratcheting pre...." European Journal of Mechanics, A/Solids 114: 105770

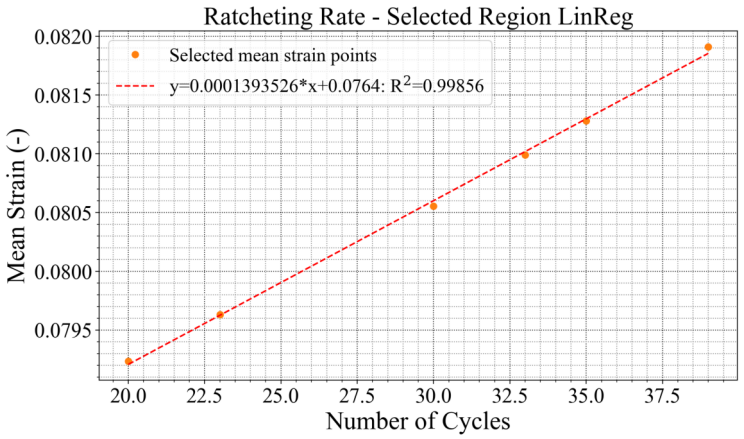
DOI: https://doi.org/10.1016/j.euromechsol.2025.105770

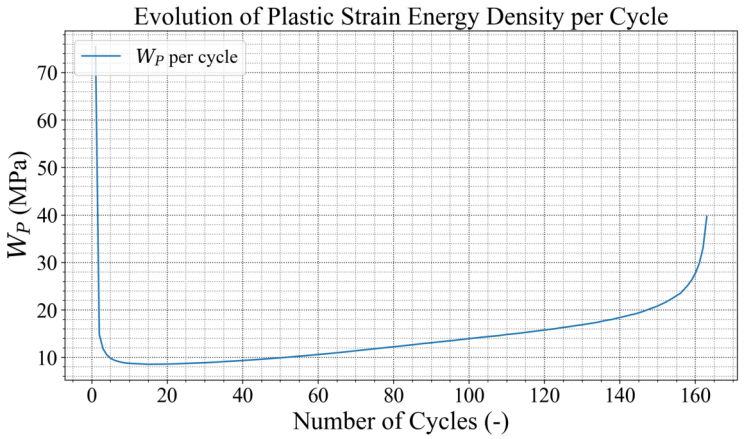












Additional values:

W_p^{tot} (MPa)	ΔW_p^{mean} (MPa)	ΔW_p (MPa)	ΔW_p^{VarE} (MPa)
1228.44	16.379	12.601	12.539